
Mechanical properties of the dentinoenamel junction: AFM studies of nanohardness, elastic modulus, and fracture

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Abstract: The dentinoenamel junction (DEJ) is a complex and poorly defined structure that unites the brittle overlying enamel with the dentin that forms the bulk of the tooth. In addition, this structure appears to confer excellent toughness and crack deflecting properties to the tooth, and has drawn considerable interest as a biomimetic model of a structure uniting dissimilar materials. This work sought to characterize the nanomechanical properties in the region of the DEJ using modified AFM based nanoindentation to determine nanohardness and elastic modulus. Lines of indentations traversing the DEJ were made at 1–2 μm intervals from the dentin to enamel along three directions on polished sagittal sections from three third molars. Nanohardness and elastic modulus rose steadily across the DEJ from bulk dentin to enamel. DEJ width was estimated by local polynomial regression fits for each sample and location of the mechani-

cal property curves for the data gradient from enamel to dentin, and gave a mean value of 11.8 μm , which did not vary significantly with intratooth location or among teeth. Nanoindentation was also used to initiate cracks in the DEJ region. In agreement with prior work, it was difficult to initiate cracks that traversed the DEJ, or to produce cracks in the dentin. The fracture toughness values for enamel of 0.6–0.9 $\text{MPa} \cdot \text{m}^{1/2}$ were in good agreement with recent micro-indentation fracture results. Our results suggest that the DEJ displays a gradient in structure and that nanoindentation methods show promise for further understanding its structure and function. © 2000 John Wiley & Sons, Inc.

Key words: dentinoenamel junction; nanomechanical properties; modulus; hardness; fracture

INTRODUCTION

The dentinoenamel or dentin-enamel junction (DEJ) is a complex and critical structure uniting the hard and brittle enamel with the tougher and softer dentin. It plays a critical role as the initiation surface for ameloblastic and odontoblastic activity during tooth formation.¹ In maturity, it appears to be critical to the biomechanical integrity of the tooth,² because it joins these structurally diverse calcified tissues. The struc-

ture of the DEJ is generally described as scalloped with its convexities directed toward the dentin and concavities directed toward the enamel.³ This structure is thought to lead to enhanced bonding between these calcified tissues. However, there is little information available on the size and variations of these structures. There may be considerable variability between animals, individual teeth, and within a given tooth.^{4–6} Lin et al.⁷ found that the scalloped structure contained microscallop as well as collagen type I fibrils that appeared to emanate from the dentin and coalesce to form coarse fibrils that crossed the DEJ and inserted into the enamel. In addition, the available published micrographs show finer structures that have not as yet been defined. The DEJ appears to form a complex interface with at least three levels of microstructure: scallops of varying size that may vary with location; microscallop housed within each scallop; and a finer nanolevel structure within each microscallop. As pointed out by Lin et al.,⁷ the DEJ may be a microstructurally distinct and mechanically tougher

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portion of the tooth that is essential to its function and acts to prevent enamel cracks from propagating across the interface, thus preventing catastrophic tooth fractures. Recently, Pioch and Staehle⁸ examined the shear strength of human and bovine teeth in the region of the DEJ and reported that no fractures occurred exactly at the DEJ, indicating the toughness of the DEJ and the difficulty associated with accurate measurements of its properties.

These difficulties suggest the need for additional methods of evaluating the DEJ. One approach is to use fracture mechanics rather than simple tests of tensile or shear strength. Rasmussen et al.⁹ used a work-of-fracture (W_f) approach to determine the fracture characteristics of enamel and dentin. They found anisotropic W_f values for enamel and dentin and also applied similar methods to evaluate the DEJ region, for which a value of about 336 J/m² was found in comparison to 221 and 391 J/m² for fracture perpendicular and parallel to the dentin tubules, respectively.¹⁰ They noted that minimal failure actually occurred at the DEJ, and concluded that the work of fracture of the dentin increases near the DEJ.

Lin¹¹ and Lin and Douglas² conducted studies of the fracture toughness of the DEJ and suggested that the DEJ appears to have wider functional width, of 50–100 μ m, than anatomic appearance; probably undergoes plastic deformation during crack propagation, and most likely serves as a crack deflector and blunter to

prevent catastrophic tooth or enamel fractures. They reported substantially higher values of fracture toughness, K_{IC} , and fracture energy, G_C , for the bovine DEJ than for bovine enamel or bovine dentin (see Table I for literature values of mechanical properties^{2,9–24}). These studies support the idea that the DEJ has important and unique characteristics. Many questions remain about its properties, because of its small size and the difficulties of making accurate mechanical measurements.

Xu et al.²⁴ used microindentation to evaluate the hardness, modulus, and toughness of enamel and dentin and areas close to the DEJ. They reported values of about 3.0–3.6 GPa for enamel hardness, 80–94 GPa for enamel modulus, and toughness of 0.5–1.3 MPa $\cdot$ m^{1/2} that depended on tooth and orientation of the enamel. For dentin, hardness values of 0.57 GPa and an elastic modulus of 20 GPa were reported, but they were unable to introduce fractures in the dentin. White et al.²⁵ reported on the microhardness gradient across the DEJ and found a broad region about 100 μ m wide in which the properties decreased from enamel to dentin, which appeared to be much wider than the optical appearance of the DEJ.

Recently, advances in nanoindentation measurements have been applied to enamel and dentin.^{16,19} Nanoindentation has the advantage of using a much smaller indenting stylus than conventional microhardness testers, so it is possible to sample material prop-

TABLE I
Mechanical Properties of Enamel, Dentin, and DEJ^a

	Enamel	Dentin	Peritubular Dentin	Intertubular Dentin	DEJ
Compressive Strength (MPa)	95–140	230–370			
Young's Modulus (GPa)	9–90	11–20	31	16.3	
Shear Strength (MPa)	90.2	36–138			
Tensile Strength (MPa)	8–35	31–104			
Microhardness (GPa)	3.2–4.4	0.25–0.8			
Nanohardness	3.1–3.4		1.6–2.5	0.13–0.58	
Work of Fracture (J/m ²)	200 ^b 13 ^c 1.1 ^b	270 ^b 550 ^c 2.6 ^b			336
K_{IC}	0.52 ^c	2.8–3.1 ^c			
K_{IC} (MPa $\cdot$ m ^{1/2})	0.52 ^c 1.3 ^b	2.8–3.1 ^c 2.8 ^b			3.38 ^b
G_{IC} ^{560^b} (J/m ²)	23 ^d	669–742 ^c 750 ^d			988 ^b 695 ^c

^aCompiled from Bowen & Rodriguez;¹² Braden;¹³ Waters;¹⁴ Duck;¹⁵ Van Meerbeek et al.;¹⁶ Craig & Peyton;¹⁷ Sano et al.;¹⁸ Willems et al.;¹⁹ Kinney et al.;^{20–21} Watanabe et al.;²² Rasmussen et al.;⁹ Rasmussen;¹⁰ El Mowafy and Watts;²³ Lin & Douglas;² Lin;¹¹ Xu et al.²⁴

^bPerpendicular to prisms or tubules.

^cParallel to prisms or tubules.

^dBovine.

erties with smaller applied loads and finer spatial resolution. Furthermore, from the load displacement behavior, it is possible to determine Young's modulus for the material.²⁶ In addition to static behavior, the dynamic response of materials (e.g., relaxation time) can be determined.²⁷ However, conventional nanoindentation does not allow precise positioning of indentations and, thus, aligning the indenter tip with the feature of interest is difficult. To overcome this disadvantage, the atomic force microscope (AFM) was first modified to simultaneously image topography and stiffness, while allowing site-specific indentations to be made before or after imaging. This allowed the first site-specific hardness values of fully hydrated intertubular and peritubular dentin to be determined with values of 2.3 and 0.5 GPa, respectively, as compared to a value of 3.1 GPa for enamel.^{20–21} Recent modifications to the AFM allow sequential AFM imaging and provide load displacement curves at selected sites with loads as low as 1 μN while detecting displacements of up to 35 μm . This method has been used to evaluate the hardness and modulus for fully demineralized dentin,²⁷ as well as measurement across the resin-dentin hybrid zone used in dentin bonding.²⁸

The DEJ is a complex region of small size and irregular geometry, which probably forms a graded interphase that makes study with conventional mechanical testing methods especially difficult. Macroscopic tensile, compression or shear tests are difficult because of the complex geometry that results in imprecise and nonuniform stress distributions, as well as the small size of human teeth. The purpose of this work was to measure the nanohardness and elastic modulus characteristics of a portion of the human DEJ using modified AFM methods, and to apply a new approach to generating precise fractures to define the unique properties of the DEJ.

MATERIALS AND METHODS

Sample preparation

Third molars were extracted from research subjects requiring such extractions as part of their dental treatment. All subjects enrolled in this research responded to an informed consent protocol that has been approved by the UCSF Institutional Committee on Human Research. The extracted teeth were sterilized by gamma radiation²⁹ and stored in purified water at 4°C until prepared. Three teeth were sagittally sectioned to prepare 1 mm thick dentin slabs containing enamel, the DEJ, and dentin. The surface of each slab was metallographically polished through a series of SiC abrasive papers down to 4000 mesh. Subsequently, they were polished with diamond pastes of 1 and 0.25 μm on a hard cloth and finally, finished by polishing with 0.05 μm alumina powder slurry. The samples were ultrasonically cleaned in

distilled water at the end of each polishing step to remove polishing agents and remnant smear layer. Similar procedures were carried out to prepare five samples from slabs cut parallel to the occlusal surface, after fracture experiments failed to produce cracking in samples prepared from the sagittal sections.

AFM imaging and nanomechanical properties

The AFM used in this study was a Nanoscope III (Digital Instruments, Santa Barbara, CA) with the standard head replaced with a Triboscope indenter system (Hysitron Inc., Minneapolis, MN), as previously described.²⁷ In this configuration the standard AFM head is replaced by a capacitive sensor. The sensor consists of two fixed outer electrodes (drive plates) that are driven by AC signals 180 degrees out of phase relative to each other. Due to the small spacing between two plates, the electric field changes linearly from one to the other. Therefore, the electric field potential is maximum at the drive plates and zero at the center between the two plates. The center, or pickup electrode, is suspended in a manner that allows it to move up and down in the region between the two drive plates. The pickup electrode assumes the electric potential of the space between the two drive plates. This results in a bipolar output signal that is equal in magnitude to the input signal at the maximum deflection, and zero at the center position. The synchronous detector converts the phase and amplitude information from the sensor output into a bipolar DC output signal. The output signal is actually a reading of the pickup electrode position. In the imaging mode, this signal is used as a feedback to the piezoceramic tube for constant force contact imaging. In the indentation mode, the feedback is cut off and a voltage ramp is applied to the lower drive plate. As a result, an electrostatic force is generated between the pickup electrode and the drive plate. The force can be described as

$$F = k_e V^2, \quad (1)$$

where k_e is the electrostatic force constant and V is the applied voltage. The voltage ramps are formulated to produce triangular, trapezoidal, or square force-loading profiles of the sample. For experiments in air, the force is applied to the sample through a diamond tip glued to a tapped polymer holder attached to a pickup electrode by a small screw. In liquid, the tip is glued to a tungsten rod with a large aspect ratio, which in turn is attached to the polymer holder. In this configuration, the diamond tip and portion of the tungsten rod are immersed in water and, as a result, the meniscus force remains constant as the height of liquid changes due to vaporization. This force can be easily accounted for prior to contacting the sample. In the imaging mode, the minimum contact force applied to the sample is about 1 μN . In the indentation mode, loads up to 30,000 μN can be applied. A cube corner indenter with a tip radius of curvature of about 20 nm was used for both imaging and indentation. All images and indentations for modulus and nanohardness across the DEJ region were made with the sample submerged in water. The sharp cube corner indenter displaces more than three times the volume of the Berkovich indenter at the same

load. The cube tip is particularly useful for fracture toughness measurements, because the cracking threshold becomes lower than for the Berkovich indenter. The hardness, H , is calculated on the basis of maximum force, F_{max} , divided by the projected contact area at maximum load, a , while the elastic modulus, E , is calculated from the contact stiffness, S , defined as the slope of the linear portion of the force/displacement curve during unloading near the maximum load.^{26,30}

$$H = F_{max}/a, \quad (2)$$

$$E = \sqrt{[\pi/a] \cdot S/2}. \quad (3)$$

E and H were measured at intervals of 1 or 2 μm along lines of 50 μm length encompassing the enamel, DEJ, and dentin. Three intratooth locations used were: (1) along a central line from the incisal portion of the tooth toward the pulp chamber; (2) from the cusp tip toward pulp horn; and (3) along the central line in the mesial-distal direction. These were called locations 1, 2, and 3 for convenience.

Fracture toughness

It has been well established that the fracture patterns generated by small-scale contact events can be related to mechanical properties of ceramic materials. In particular, sharp indenters' contacts generally produce elastic/plastic stress fields that govern crack development that can be related to mechanical properties such as toughness, K_c , hardness, H , and elastic modulus, E .^{31–33}

Cracks produced by indentation contact were classified by Lawn et al.³³ into two primary systems: those cracks that form symmetry median planes containing the load axis, and those that form laterally on planes closely parallel to the sample surface. In the present work, only the former are considered, because the formulation for this system has been developed quantitatively.

Historically, Palmqvist³⁴ was the first to demonstrate that the radial cracks could be related empirically to toughness. A more fundamental approach to the median/radial problem based on Griffith/Irwin fracture mechanics was developed by Lawn and Wilshaw³¹ to include material parameters based on elastic theory that provided some indication of the role of plasticity in crack propagation. Finally, Lawn et al.³² examined the elastic/plastic force field responsible for crack extension in detail and established a rationale for pre-determining the mechanics of the median/radial system for any material whose parameters K_c , H , and E are specified.

Figure 1 demonstrates the basic components involved in the formulation of fracture toughness. At the critical indentation load, P , a median subsurface crack is formed under the indentation tip due to generation of tensile stress. The number of median cracks depends on the indenter used: four for Vickers or three for three-sided cube corners, which form parallel to the indentation diagonals. On unloading, the median cracks become unstable and radial cracks form in response to the residual tensile stress field surrounding the indenter. The stress-intensity factor, K_R , from the field

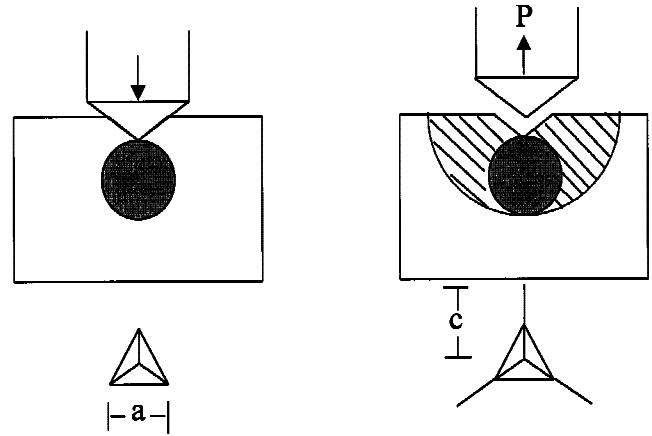


Figure 1. Schematic diagram of indentation. At the critical indentation load, P (left), a median subsurface crack is formed under the indentation tip due to generation of tensile stress and is the characteristic size for the 3-sided indenter. On unloading (right), the median cracks become unstable and radial cracks form of length C in response to the residual tensile stress field surrounding the indenter.

can be calculated from the indentation load, P_{ind} , and half surface-crack length, c , by

$$K_R = \frac{\chi P_{ind}}{c^{3/2}}, \quad (4)$$

where χ is a material constant dependent on elastic modulus, E , and hardness, H , given by:

$$\chi = \alpha \left(\frac{E}{H} \right)^{1/2}, \quad (5)$$

where α is a material-independent constant related to indentation geometry. The toughness is then calculated from the above equations at $K_c = K_R$ by denoting c as the equilibrium surface length of the post-indentation crack.

The empirical value of α is determined by the slope of $K_c(H/E)^{1/2}c^{3/2}$ vs. load for a known material. The values of 0.016³⁵ and 0.0535³⁶ are reported for Vickers and cube corner indenters, respectively. Table II lists typical values reported for engineering materials using this method.^{35–37}

AFM-based indentation is limited to the maximum load of 30,000 μN . Due to this limitation, the cube corner indenter was chosen for this work. This indenter displaces more volume than Berkovich or Vickers indenters at the same load. Consequently, the crack formation is greater with the cube corner, thereby reducing the cracking threshold. The fracture experiments were carried out in air following the hardness and modulus measurements made across the DEJ. However, the mechanical properties do not change appreciably for enamel, and dentin E and H values are only 10–20% lower when measurements are performed wet. Because there is some dependence of properties on immersion time,⁴¹ it seemed prudent to do the fracture measurements in air.

TABLE II
Mechanical Properties and Fracture Toughness Values Obtained by Indentation Technique for Some Selected Brittle Materials^a

	Soda-Lime Glass ³³	Silicon ³³	SiC Hard Coating ³⁵	Polycrystalline Graphite ³⁷
Hardness, H	5.5 (GPa)	9 (GPa)	29 (GPa)	50–120 (DPH)
Elastic Modulus, E (GPa)	70	168	330	4–12
Fracture Toughness K_{IC} (MPa-m ^{1/2})	0.75	0.7	2.97	1.5

^a(Data for soda lime glass and Si from Lawn et al.,³³ SiC from Anstis et al.,³⁵ and graphite from Ritchie et al.³⁷)

Statistical methods

To obtain estimates of the functional DEJ width, local polynomial regression (loess) with 95% confidence bounds³⁸ was fitted for each sample and location of the mechanical property curves for the data gradient from enamel to the DEJ and into dentin. Loess is a nonparametric, minimal assumption smoother (proc loess, SAS Institute, Cary, NC). (Linear local fitting and a smoothing parameter of 0.175 were used.) The DEJ width was then determined from the intersection of the transition data confidence bounds with the values for the enamel and dentin; the functional DEJ width was estimated as the mean from hardness and modulus derived widths. Bootstrap resampling³⁹ with 1000 resamples (S-PLUS, MathSoft, Seattle, WA) was used to estimate mean width and lower and upper 2.5% empirical percentiles for the independent samples. An exact Friedman's test was used to compare DEJ widths among locations within samples (StatXact, Cytel, Inc, Cambridge, MA). Exact 2-factor repeated measures (mixed) models⁴⁰ were used to assess differences among locations within samples for 13 serial measurements 2 μm apart near the DEJ; the location $\times$ distance interaction was also assessed (XPro, X-Techniques, Inc, Millington, NJ).

RESULTS

Nanohardness and elastic modulus

Figure 2 shows typical values of E and H across the DEJ at location 1 along with the AFM image showing the indentations. The values of E and H in enamel and dentin were essentially the same as the bulk tissues at a distance of about 10 μm from the DEJ region ($E = 20$ GPa and $H = 0.7$ GPa for dentin and $E = 65$ GPa and $H = 3.5$ GPa for enamel). Within the DEJ region both E and H increased from bulk values of dentin to those of enamel. Similar results were observed for the other locations, as shown in Figure 3 (locations 2 and 3), with slight variation in the estimated width of the transition zone associated with the DEJ. Figure 3 includes the data from all 3 teeth and 3 locations in each tooth, with 95% confidence intervals. Mean functional

DEJ width from the loess models was 11.8 μm , with 2.5% and 97.5% bootstrap percentiles: 10.3–12.7 μm . The variations of E and H for the three locations, averaged over three samples are shown in Table III. No significant differences, using the exact Friedman's test, were determined among locations ($p = 0.944$). No significant differences among locations ($p = 0.324$) or in the distance pattern among locations (i.e., location $\times$ distance; $p = 0.995$) were detected with the exact mixed models.

Fracture toughness

No clearly observable cracks were initiated in dentin up to the 30,000 μN limit of the instrument. However, at these indentation loads, there was considerable material displaced and piled up, which made visualization of cracking difficult, and any contorted cracks that might have been present cannot be used for calculating fracture toughness. On the other hand, cracks were consistently observed on the enamel side near the DEJ with the maximum load ranging between 6000–8000 μN in sections prepared parallel to the occlusal surface. The cracks appeared to be independent

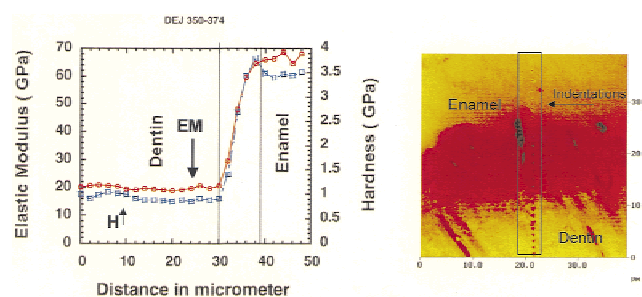


Figure 2. AFM image of indentations across DEJ near the center of the occlusal portion of the DEJ. Curves show the corresponding hardness and elastic modulus determined from the indentations. Estimates of the DEJ based on the width of the transition in mechanical properties were estimated by local polynomial regression fits to the data in the zone and its intersections with the data in enamel and dentin.

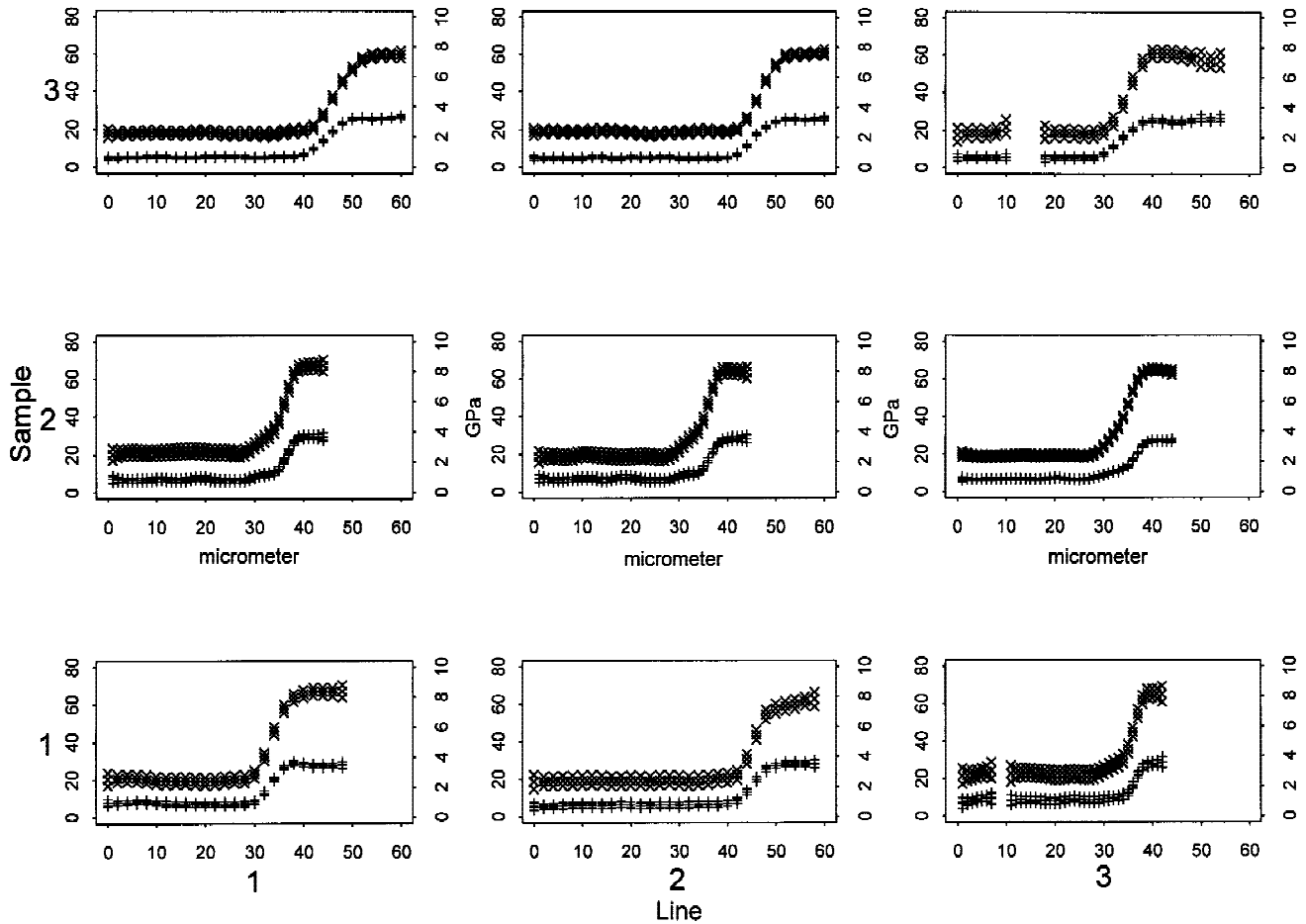


Figure 3. Modulus (GPa, vertical left axis and upper line) and hardness (GPa right vertical axis and lower line) across enamel-DEJ-dentin interface (μm) for each sample (horizontally) and line (vertically) with loess smoothed predictions and 95% confidence bounds.

of the indentation direction. The load was reduced to avoid the material pile up, and no cracks were initiated when samples were prepared in the sagittal orientation up to a load of 14,000 μN , which did not give excessive material pile-up for this device. Figure 4 shows typical indentation-induced cracking near the DEJ in an occlusal section with a maximum applied load of 6000 μN . Averaging the E and H of enamel from the local indentation measurements gave estimated fracture toughness of $K_{\text{Ic}} = 0.6\text{--}0.9 \text{ MPa} \cdot \text{m}^{1/2}$ for the teeth tested in the occlusal orientation. It should be noted that not all the crack-induced indentations resulted in cracks lying in directions parallel to

TABLE III
Young's Modulus and Hardness Values from Nanoindentation for Enamel and Intertubular Dentin Adjacent to the DEJ with [95% Confidence Intervals]

	Enamel	Dentin
Modulus (GPa)	63.55 [62.09, 65.02]	19.65 [18.61, 20.69]
Hardness (GPa)	3.51 [3.38, 3.63]	0.83 [0.74, 0.92]

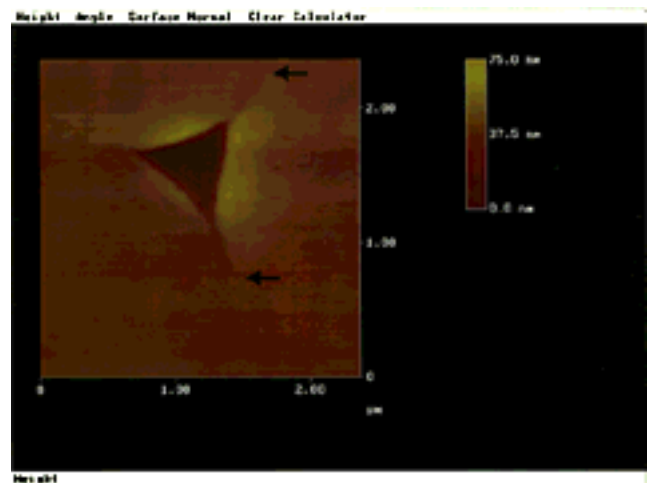


Figure 4. Cracks were initiated on enamel side near DEJ with an applied load of 6000 μN . Based on the crack length and using the local average E and H of enamel gave an estimated fracture toughness of $0.7 \text{ MPa} \cdot \text{m}^{1/2}$.

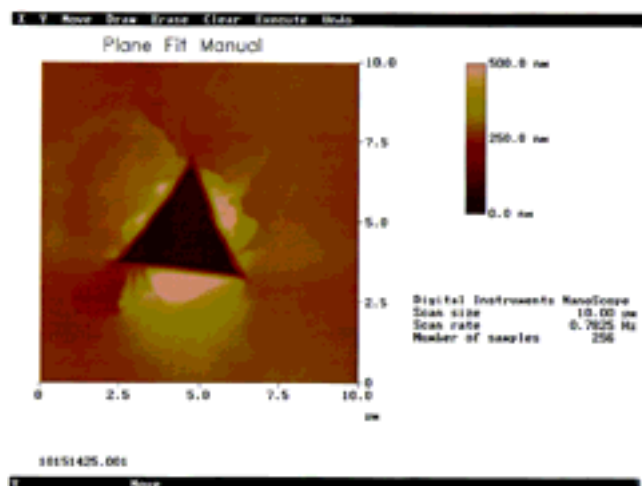


Figure 5. Indentation-induced cracks did not always lie in directions parallel to the indentation diagonals. In this sample, the crack followed a tortuous path and a portion of the enamel delaminated.

the indentation diagonals. In some cases, tortuous crack paths were observed and portions of the enamel delaminated, as shown in Figures 5 and 6.

The DEJ may be under considerable stress. This can be seen from Figure 7 in which the AFM image (left side) shows a partial crack at the interface. As a result of some deep indentations (right side) the interface crack propagated over 30 μm to release the stress along the DEJ.

DISCUSSION

This is the first report, to our knowledge, of the elastic modulus and nanohardness of the DEJ and adjacent enamel and dentin using nanoindentation. Our results for the properties of enamel and dentin agree

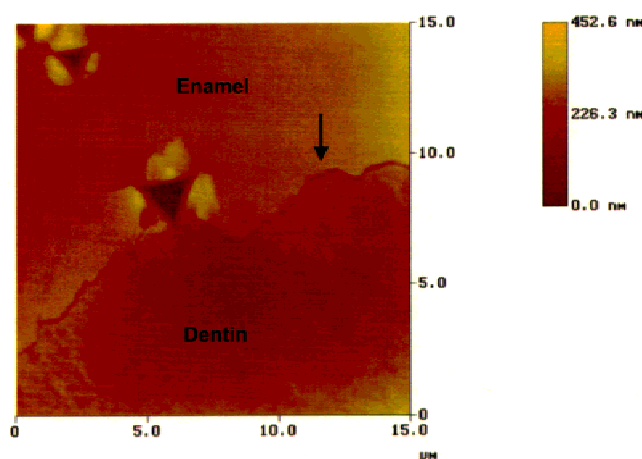


Figure 6. Another example of an irregular crack initiated near the DEJ that resulted in local enamel delamination.

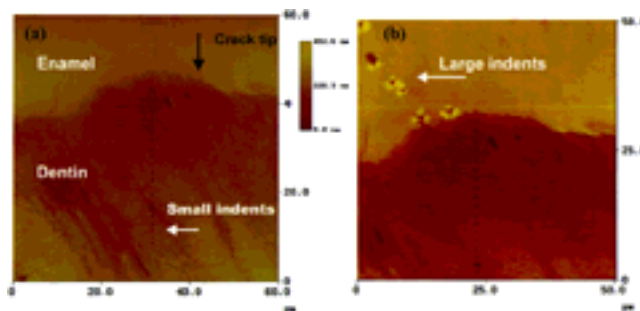


Figure 7. Frequently cracks were initiated and propagated along the enamel side of the DEJ after exposure to air. (a) A crack has progressed about 20 μm from the right edge of the image. (b) After deep indentations near the DEJ, the interface crack propagated over 30 μm to release the stress along the DEJ.

well and fall within the bounds reported in the literature, as shown in Table I. However, there is a remarkable spread in the reported values, but our results for dentin were in good agreement with measured and theoretical values of 15–16 GPa for the modulus suggested by Kinney et al.⁴¹ In addition, our dentin results agree well with values for dentin reported by Xu et al.²⁴ and hardness values from microindentation reported by White et al.,²⁵ but our results for enamel were slightly lower than the values they reported for bulk enamel in a sagittal cross-section. Our slightly lower values may reflect the fact that most of our measurements were made in close proximity to the DEJ in a fully hydrated condition.

Our results indicated that a smooth transition occurs between the harder and stiffer enamel and the softer dentin. There appeared to be a monotonic decrease from the enamel side of the DEJ to the bulk dentin. None of our measurements suggested that properties decreased from the enamel side to the dentin side in discrete stages. However, caution must be used in interpreting these results, because the small size of the junction, typically on the order of 12 μm in width, allowed only several indentations within the DEJ region along the straight line paths that were used to traverse it. Thus, if variations or substructure exist within this zone, they were not detected with the current instrumentation. Rather, the DEJ appeared to be a nearly linearly graded interphase uniting the two calcified tissues. If true, this suggests that a gradient in mineral might also exist across the interface, because the mineral component of the calcified tissue is closely related to its mechanical properties.⁴² Estimates of the microscopic and functional width are quite different. Hayashi⁴³ reported direct apposition of enamel and dentin crystals at the junction based on high resolution transmission microscopy of the rat DEJ. This suggests that, aside from the complex interdigitation of enamel and dentin at the junction, that there was tight contact between enamel and dentin. Our estimates of the

width of the DEJ interphase based on its nanomechanical properties are much larger, about 12 μm , but this is small compared to the 100 μm value, reported by White et al.²⁵ based on microindentation. The scalloped and microscalloped nature of the DEJ might contribute to the very large differences seen using these different techniques. In thin sections used in TEM, the microscopic width is sampled directly. However, in nanoindentation, a small part of the 3-dimensional scallops are sampled, and this may include contributions from the enamel and dentin phases as well as the junction itself, because the tissues are highly interdigitated. Microindentation would sample an appreciably larger volume of these tissues leading to the apparent broad width of the junction. The data presented by White et al.²⁵ shows a considerable variability in the enamel and in the region close to the DEJ, which might reflect the much larger indentations used for microhardness. Additional work is needed to further clarify these characteristics.

Our fracture results agreed with prior work that showed it is extremely difficult to initiate cracks in dentin at the DEJ or propagate cracks from enamel to dentin across the DEJ.^{10,24} Our values for toughness were in good agreement with recent values reported by Xu et al.²⁴ for sagittal sections that varied from 0.52 $\text{MPa} \cdot \text{m}^{1/2}$ parallel to 1.3 $\text{MPa} \cdot \text{m}^{1/2}$ perpendicular to the enamel rods, and were also in good agreement with the values of 0.70–0.76 $\text{MPa} \cdot \text{m}^{1/2}$ reported by Hassan et al.,⁴⁴ using microhardness indentation. This suggests that this new technique can provide an extremely powerful tool for understanding local properties of enamel, dentin, and the DEJ. It is of interest to note that the fracture toughness is dependent on both the modulus and hardness, as shown from Eq. (4). One of the advantages of the present technique is that it permits localized measurement of each of these parameters, as well as subsequent indentation to generate cracks and measure fracture toughness. This should allow future studies to reduce the experimental uncertainty in these values for biological calcified tissues, such as enamel and dentin, that may show site-specific variations in properties.

It appears that the DEJ forms a complex biomechanical structure with a unique, but poorly defined multilayered microstructure of at least 3 levels of higher organic and lower mineral content than the adjacent tissues. Our results suggest that the functional DEJ width, as determined by changes in nanohardness and elastic modulus, does not vary substantially with intratooth location. However, the available fracture characteristics indicate that the DEJ differs substantially from either enamel or dentin, and probably provides a critical link that preserves the physical integrity of the tooth. Such a junction appears to confer critical properties allowing tooth structure to function efficiently. As such it may be optimized to its function

of linking materials of substantially different properties. This suggests that the DEJ could serve as an important biomimetic model for linking other pairs of dissimilar materials, a suggestion previously made by Lin and Douglas.² There are many applications for such an optimized junction, including dental and orthopedic implants, implant coating geometry, enamel and dentin bonding systems, and restorative dental interfaces in general.

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